

Deep Neural Network Kinetic Model of Oxidative Coupling of Methane Using High Throughput Experimental Data

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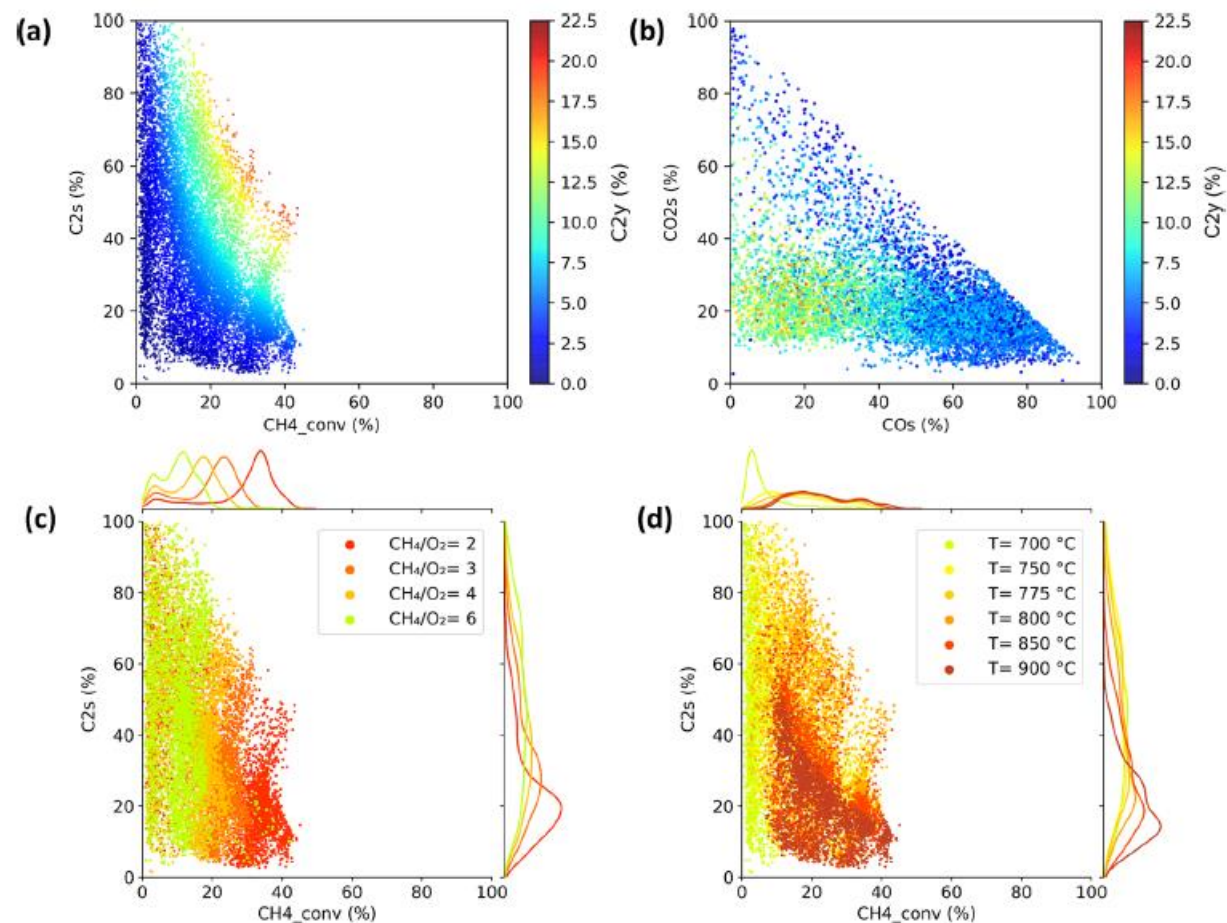
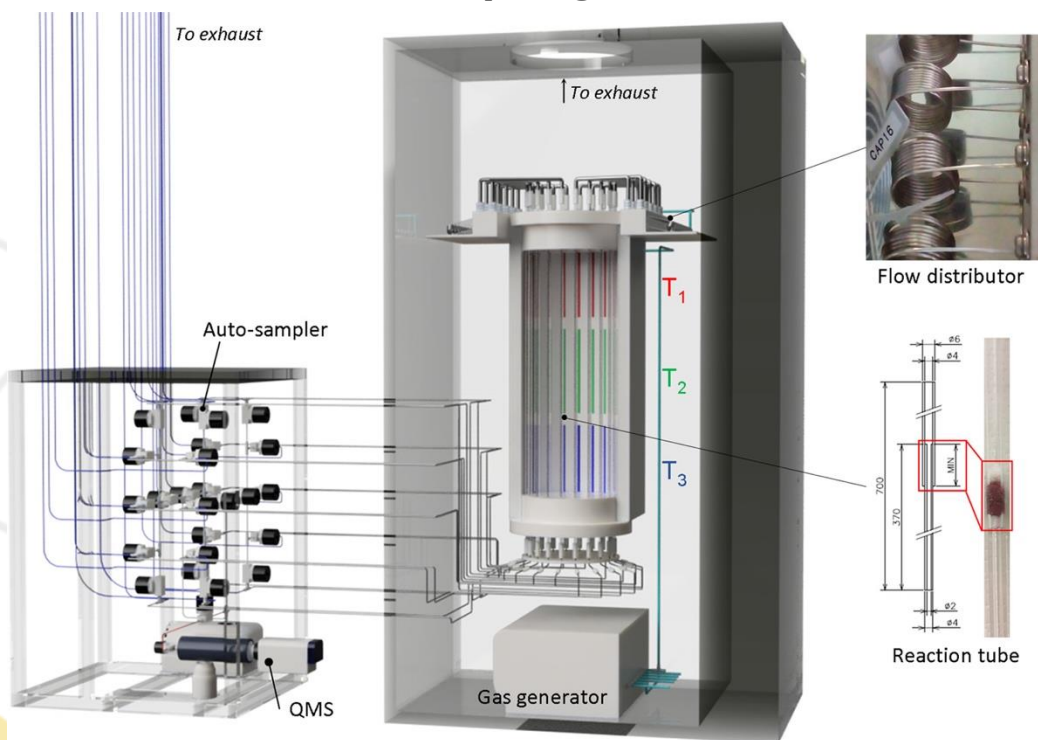
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Large catalysis datasets are becoming available

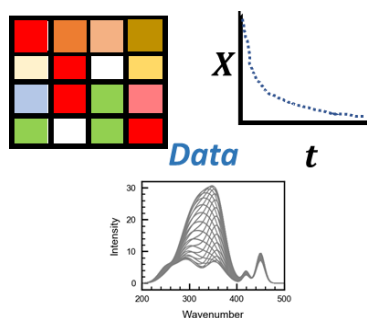
High throughput experimental set up for oxidative coupling of methane



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Such data can be used to build models

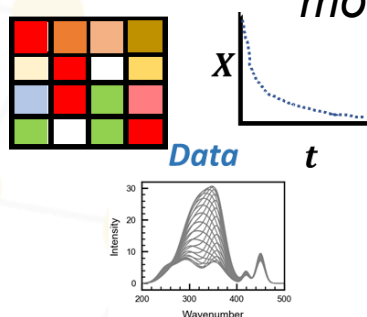
One can build black box models using these datasets



Machine learning &
data science

**Black box (NN)
models**

Incorporating domain information allows building kinetic models which are more robust to extrapolations (of catalysts and conditions)



Machine learning &
data science

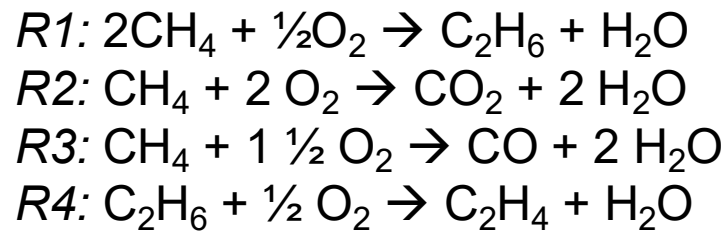
$$\frac{dX}{dt} = f(X, p)$$

**Domain-informed
kinetic models**

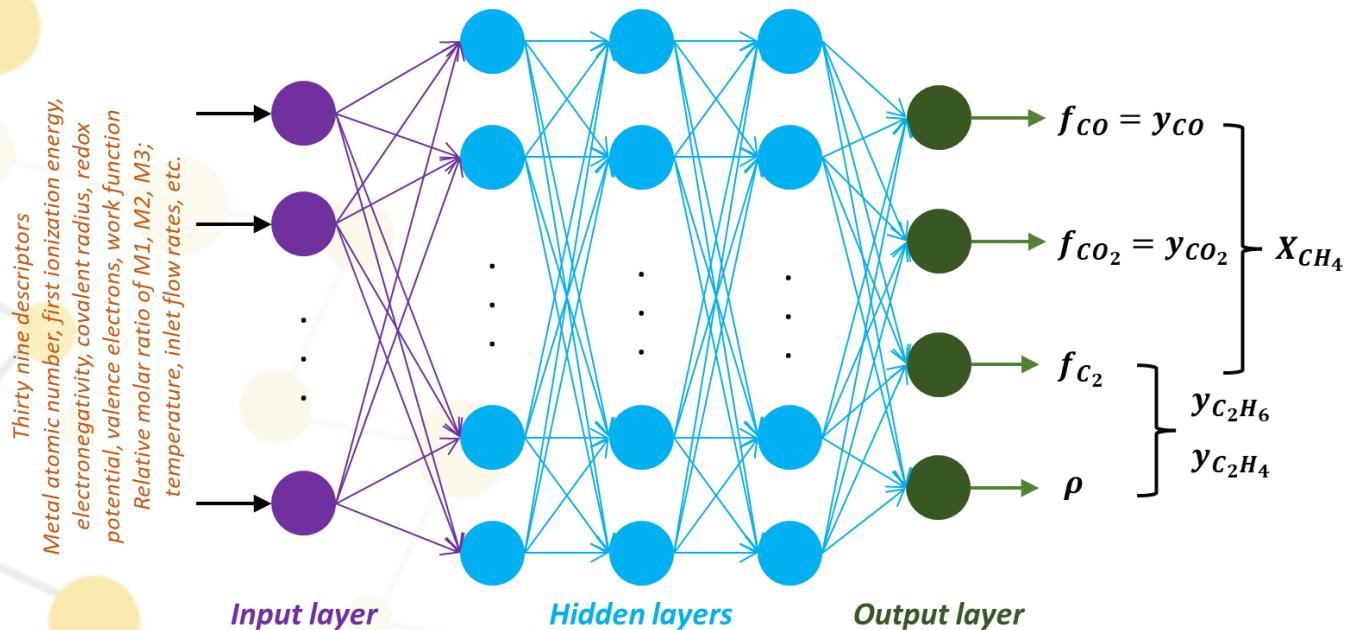
Domain constraints

Learning chemistry-informed neural network kinetic model for oxidative coupling of methane

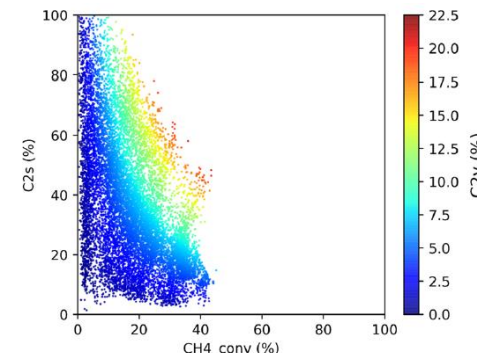
The neural network outputs are designed to ensure that the carbon balance arising from the reaction network balance is satisfied



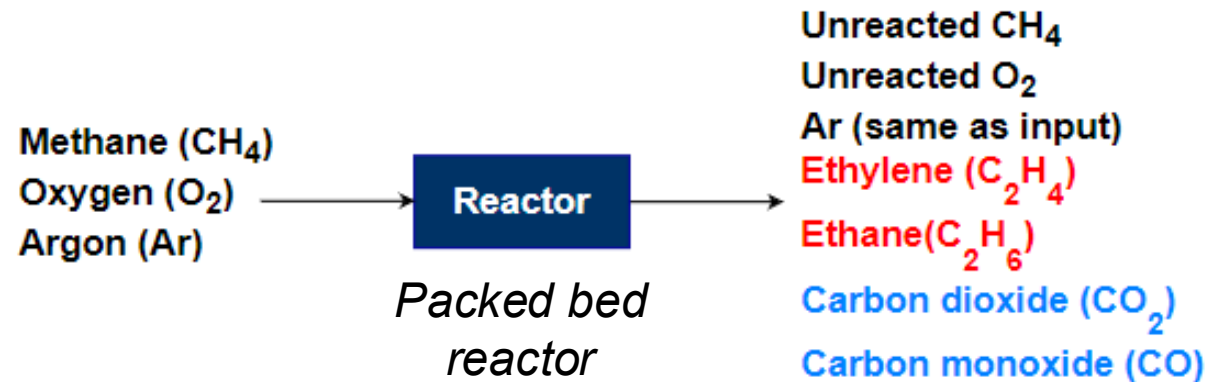
$$X_{\text{CH}_4} = y_{\text{CO}} + y_{\text{CO}_2} + y_{\text{C}_2}$$
$$y_{\text{C}_2\text{H}_6} = \frac{y_{\text{C}_2}}{\rho + 1} \quad y_{\text{C}_2\text{H}_4} = \frac{\rho y_{\text{C}_2}}{\rho + 1}$$



High throughput data



Reactor design equation for the OCM system



$$\frac{dF_i}{d\Omega} = r_i$$

Leads to:

$$F_{out,i} = F_{in,i} + \int_0^{CT} r_i F_0 d\tau$$

$$F_{out,i} \approx F_{in,i} + \sum_{j=1}^N \hat{r}_{i,j} \Delta\tau$$

F_i : Flow rate (mol/time)

r_i : species consumption/production rate (mol consumed/mol cat/time)

Ω : Catalyst amount (mol)

$\tau = \Omega/F_0$: time unit (differential residence/contact time)

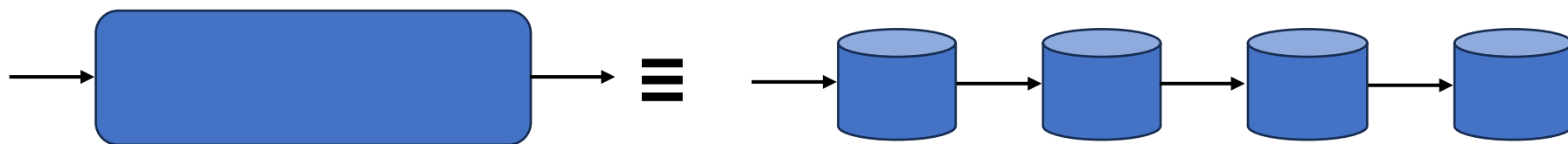
CT : Contact time (time)

$\hat{r}$: Average rate (for discretization)

$\Delta\tau = \frac{CT}{N}$: Discretization of contact time

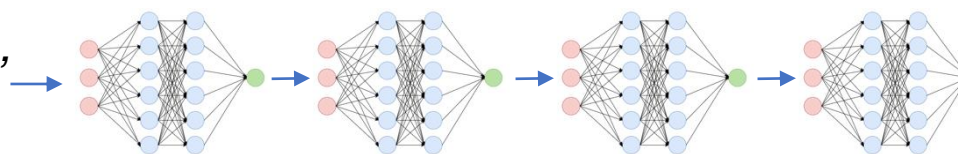
Discretization motivates the ResNet model

Packed bed reactor can be thought of as a set of consecutive CSTRs



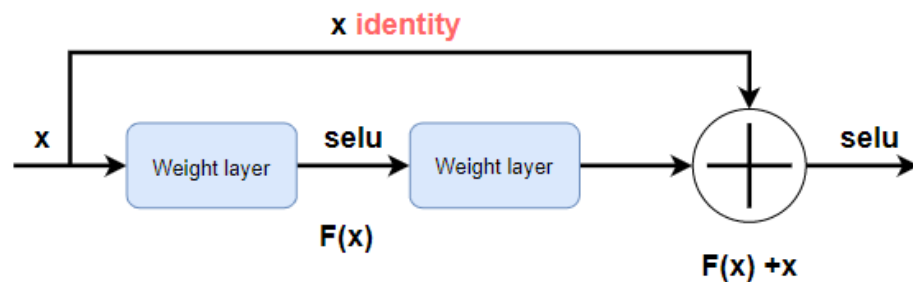
Each subunit can be represented a (same) neural network

*Catalyst, feed composition,
reaction conditions*

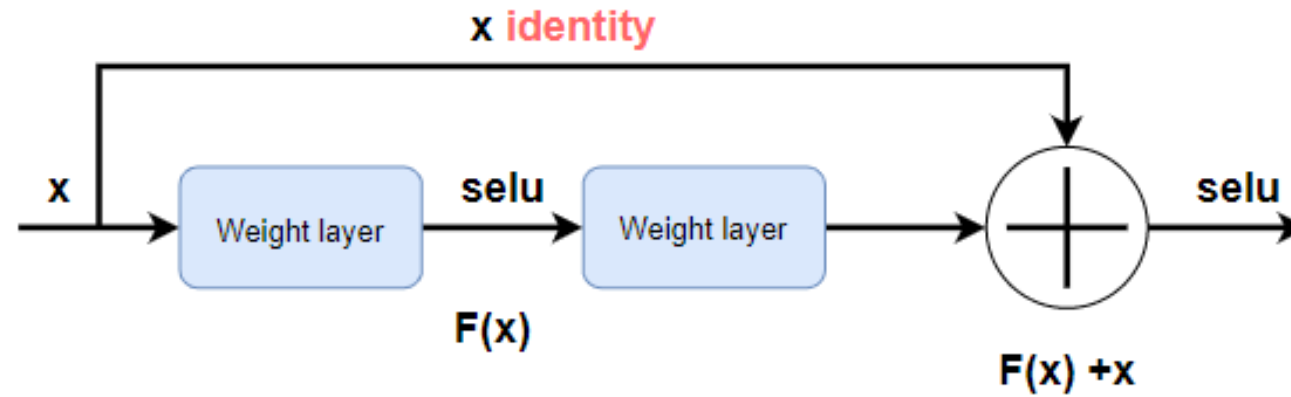


*Conversion, yield,
selectivity*

Training such deep neural networks is hard, so we assume a ResNet architecture

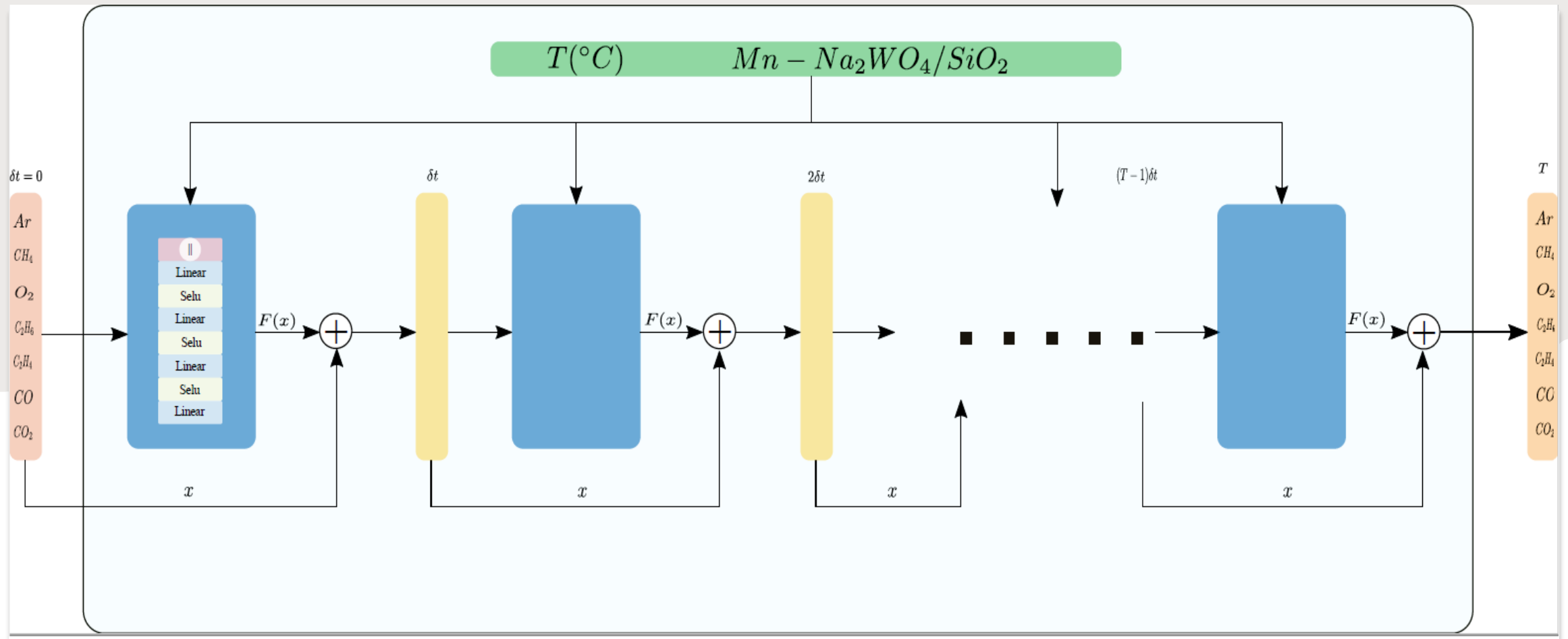


ResNet Architecture

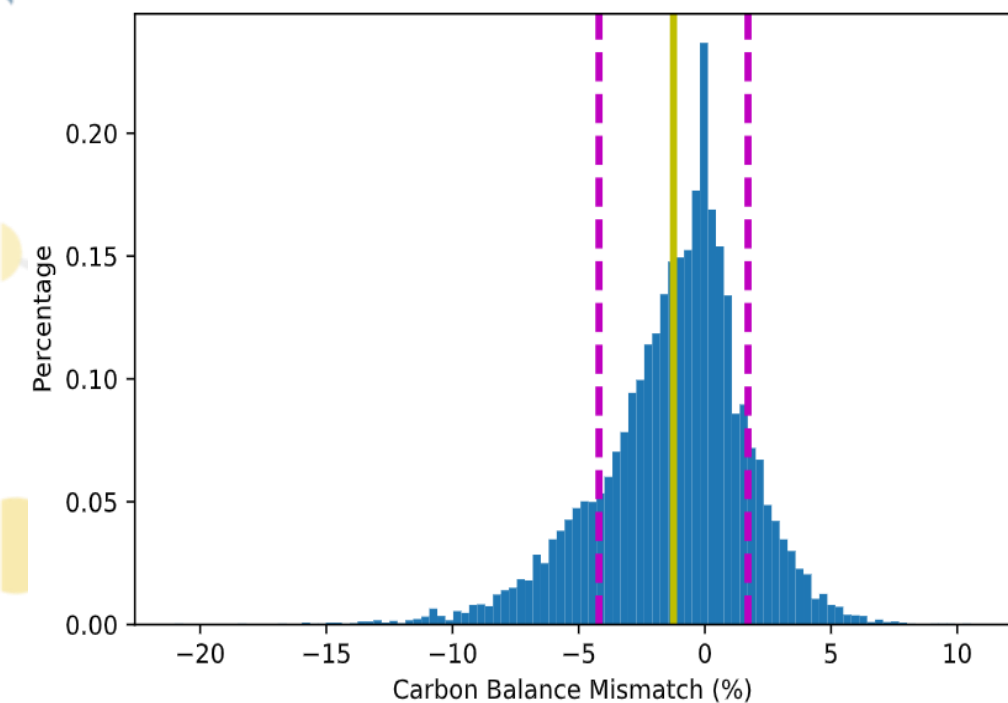


- ResNet addresses the issue of vanishing gradients
- Establishes residual connections that bypass one or more network layers
- The update of the weights is easier
- The output layer contains information from both the previous layer's input and output
- For us, $F(x)$ represents the change to the feed brought about by each reactor unit.

Architecture of Our Model

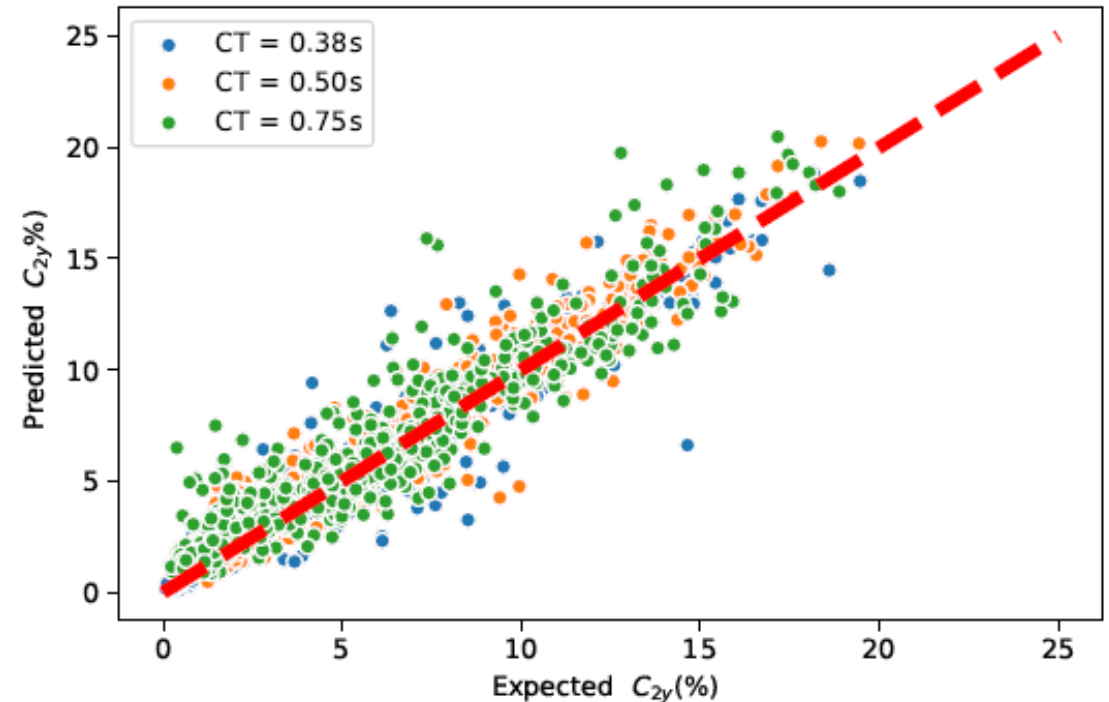
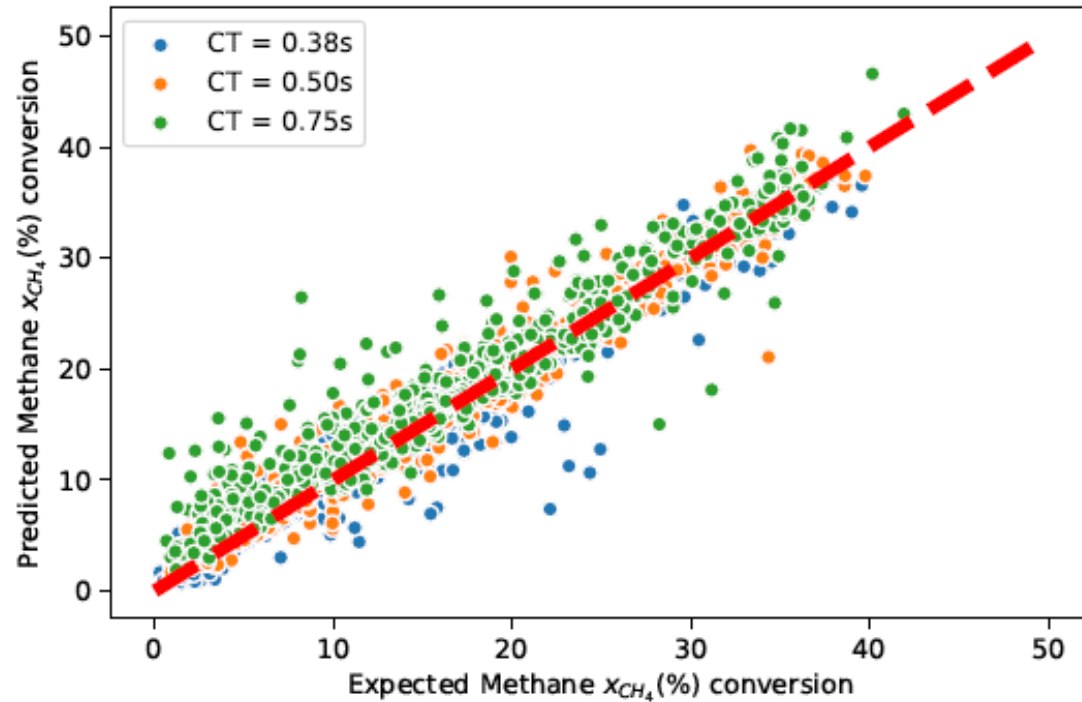


OCM Dataset

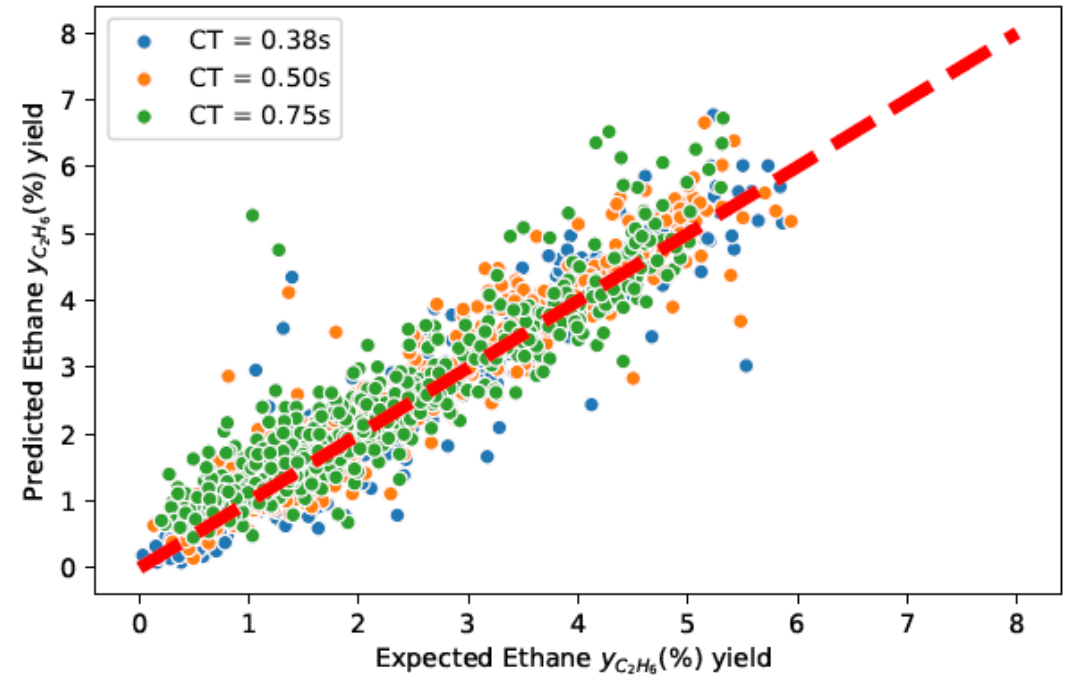
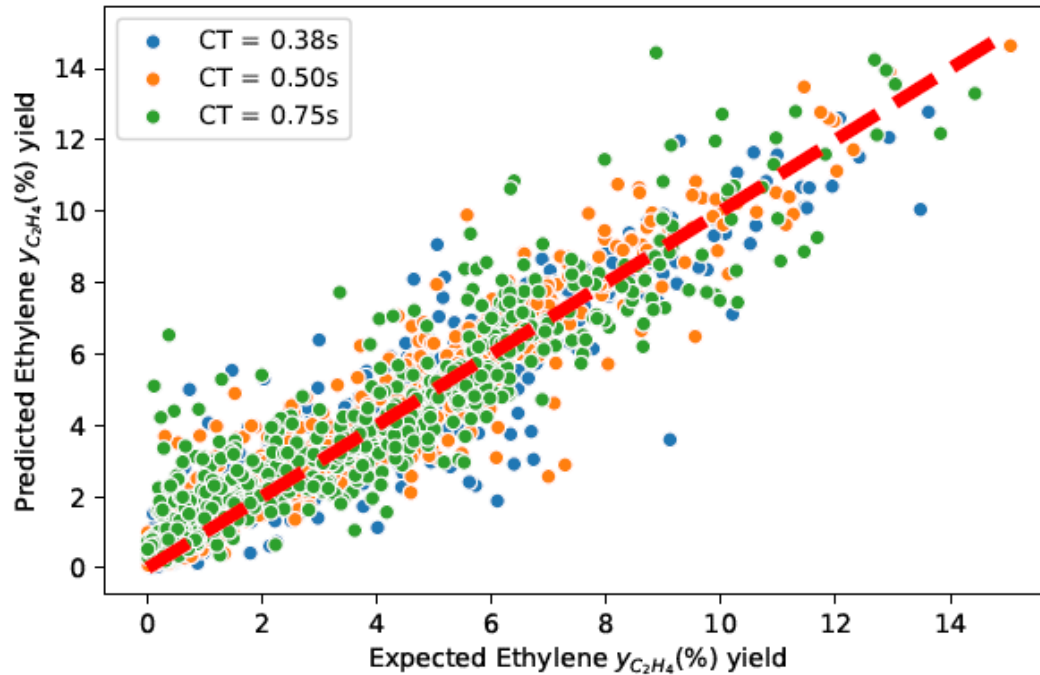


- High-throughput dataset consisting of 12,708 data points was used
- It covers 59 catalysts and support combinations with three or less metals and general formula $M_1(M_2)_{1-2}M_3O_x/\text{support}$
- Not all data points satisfies carbon and oxygen balance
- The data points are grouped by contact time (0.38, 0.5, 0.75s)
- Reaction condition`s descriptors are temperature, and argon, methane, and oxygen flow

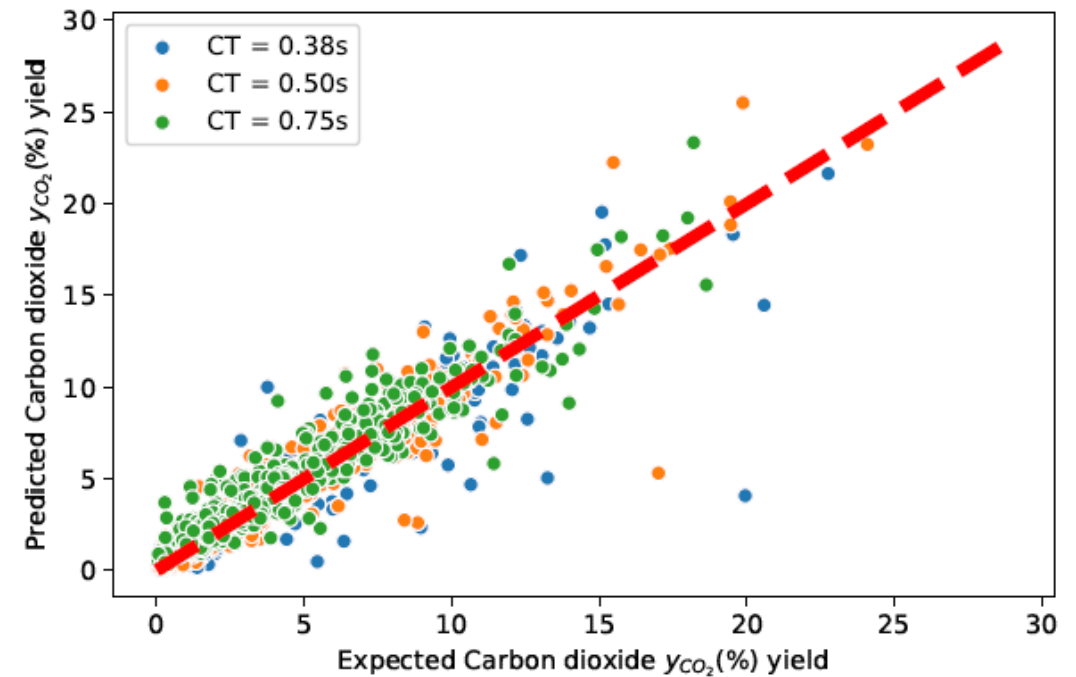
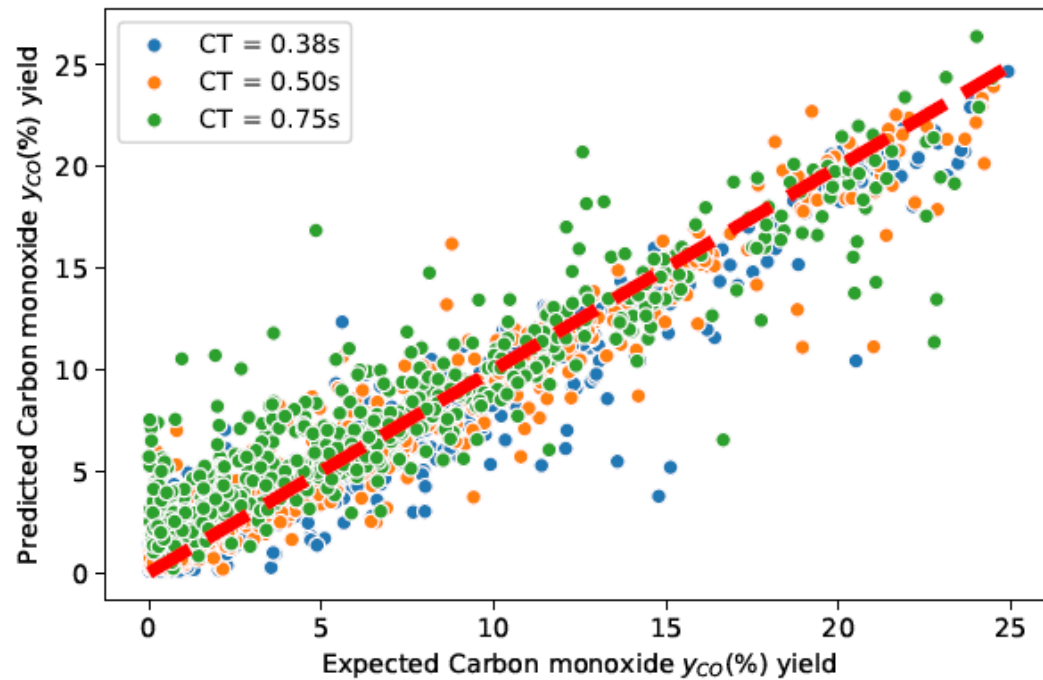
Results: Parity Plots for Methane Conversion and C_2 Yield



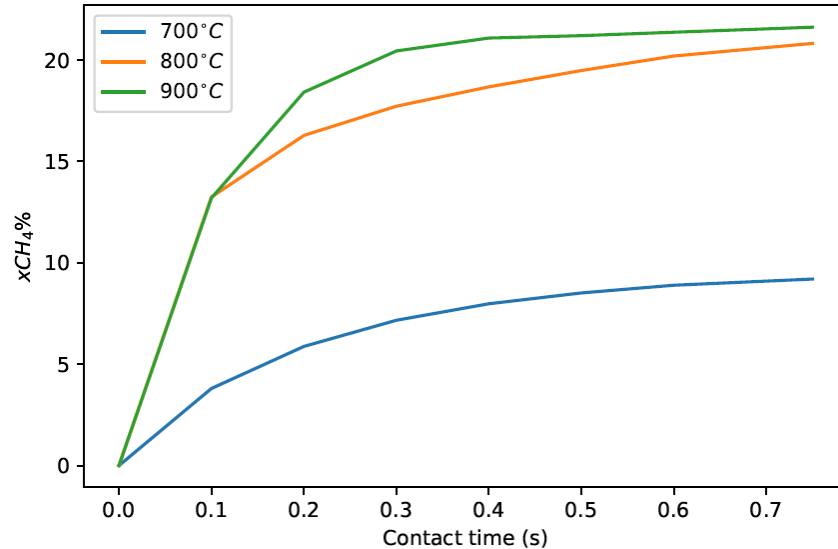
Results: Parity Plots of Ethylene and Ethane yields



Results: Parity Plots of Carbon Monoxide and Dioxide yields

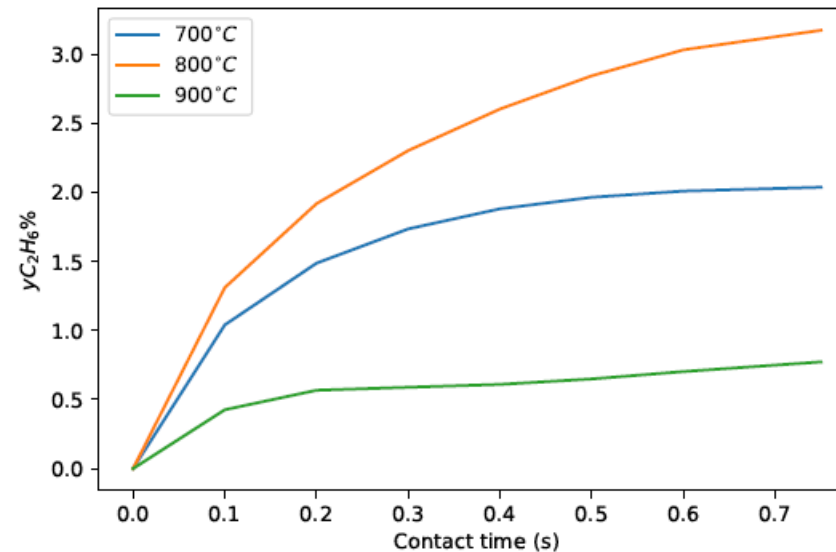
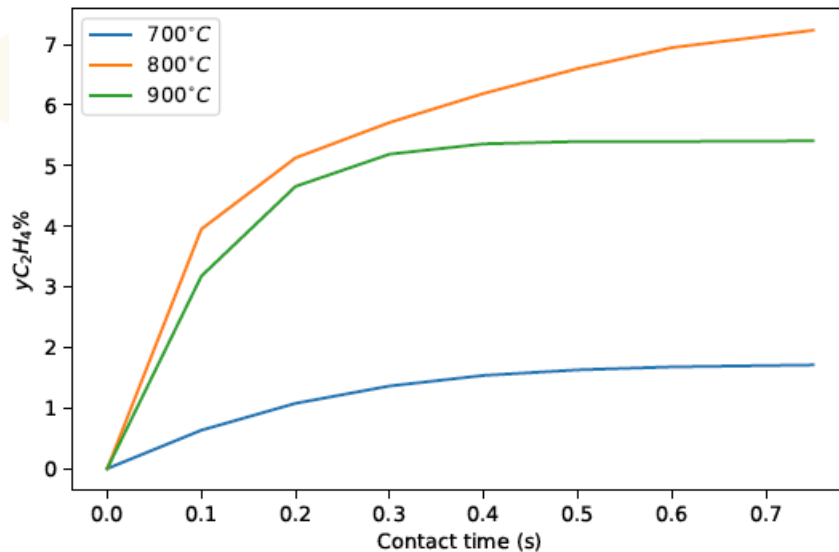


ResNet models allows us to understand the evolution of products within the reactor



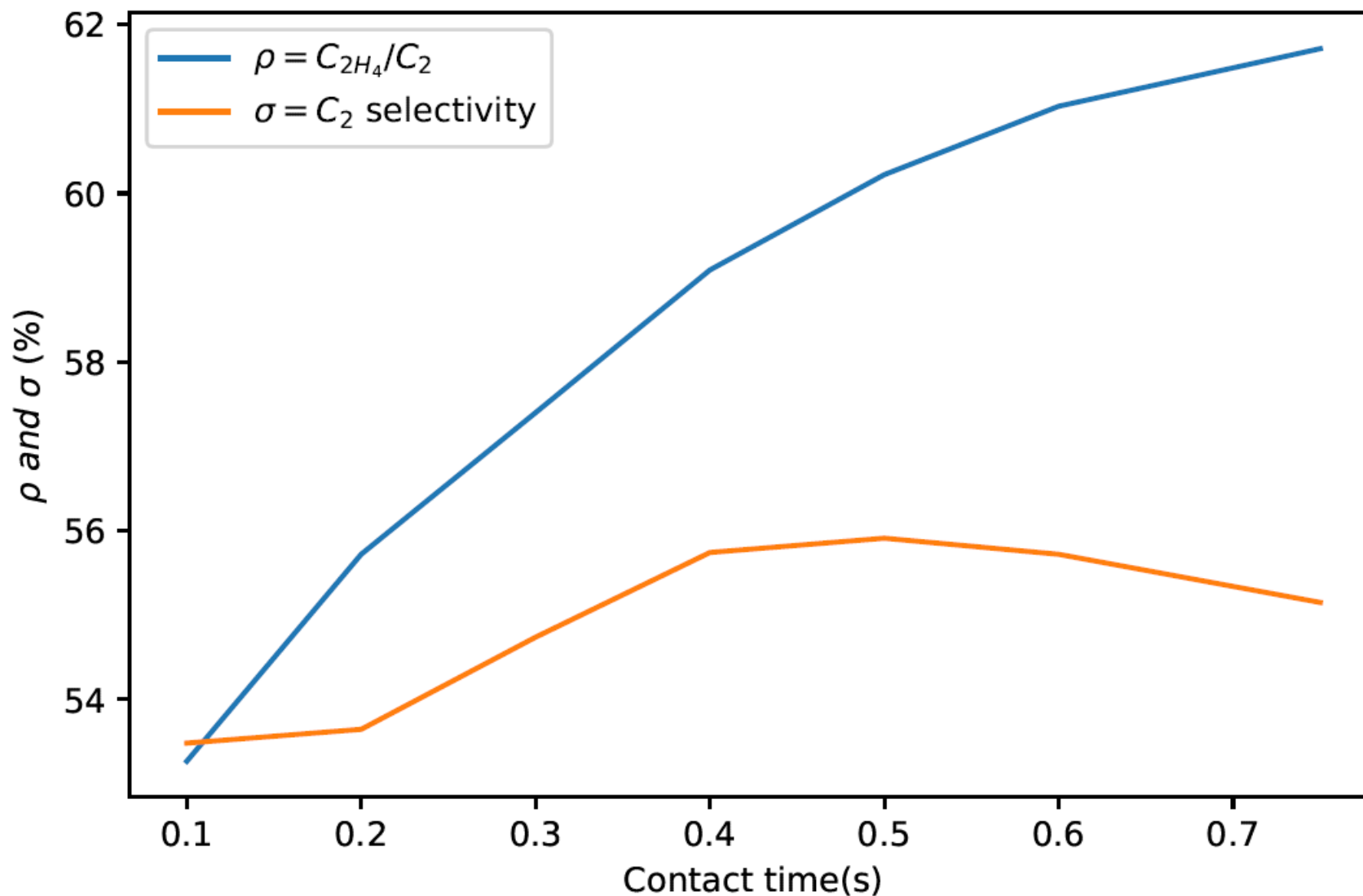
Operating Conditions:

- 4.8 ml/min methane (CH_4)
- 1.2 ml/min oxygen (O_2)
- 4 ml/min argon (Ar)
- 0.75 s contact time (CT)
- 7 residuals block for each temperature
- Mn – Na_2WO_4/SiO_2 catalyst

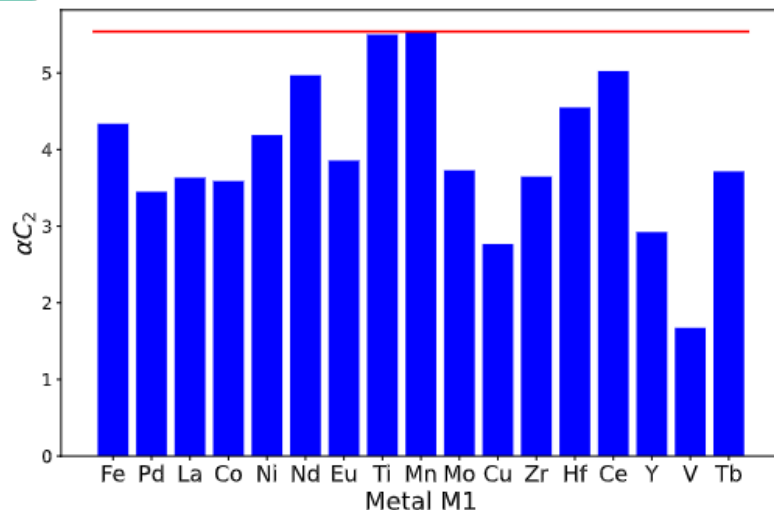


Understanding Intra-Reactor Variations

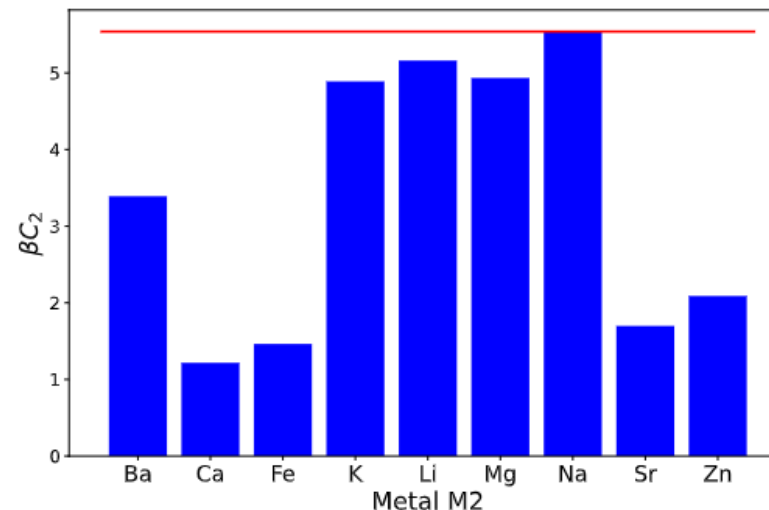
Mn – *Na*₂WO₄/SiO₂



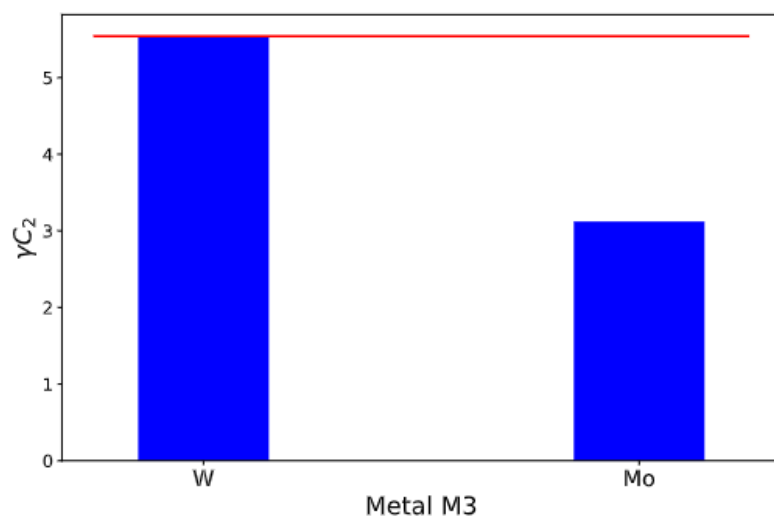
Catalyst local sensitivity analysis (C2 yield)



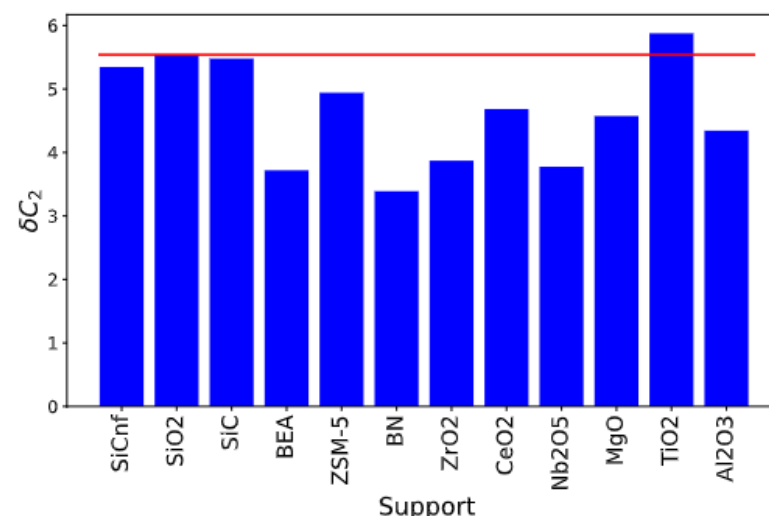
(a)



(b)



(c)



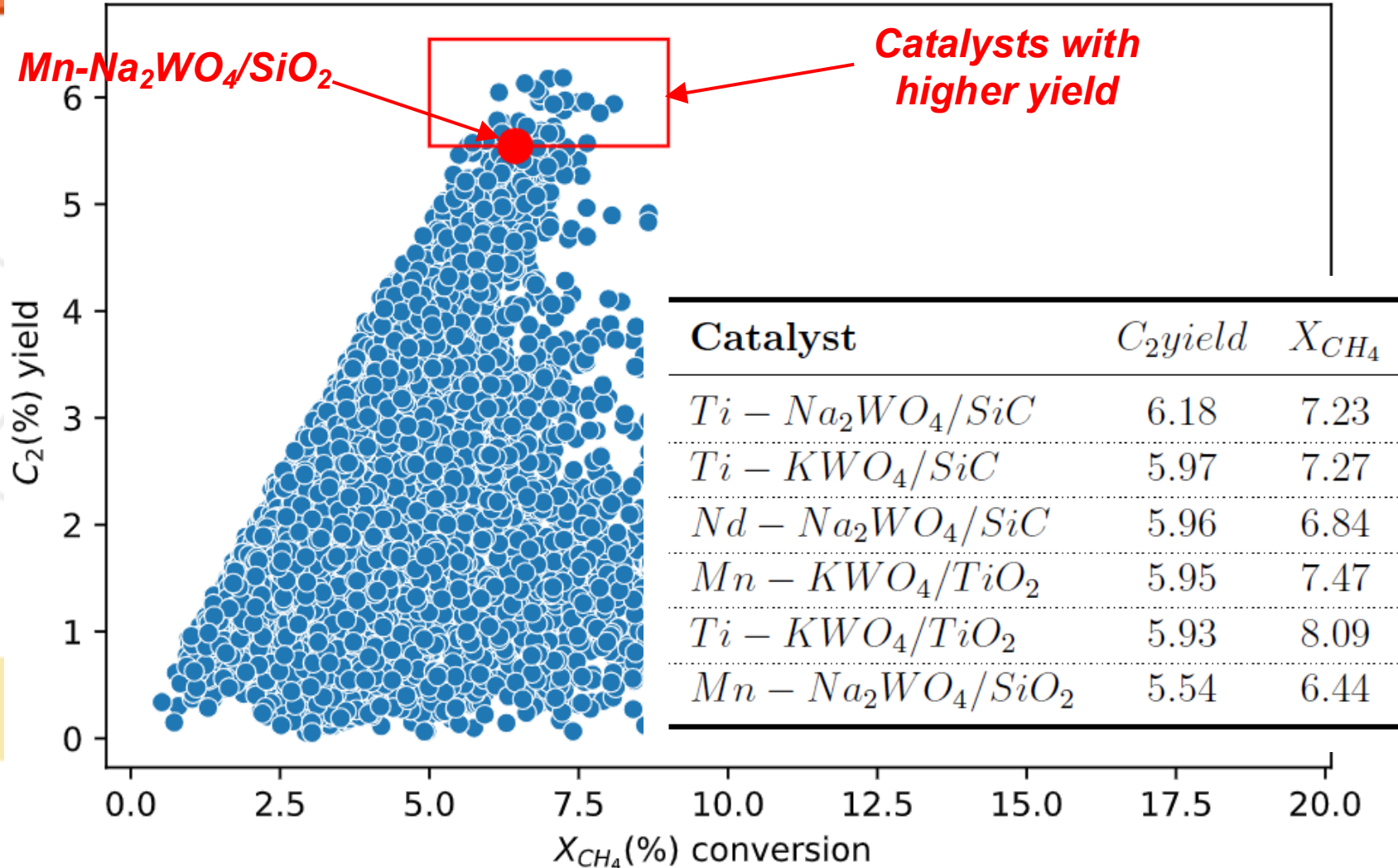
(d)

Operating Conditions:

- 10.5 ml/min methane (CH_4)
- 0.75 ml/min oxygen (O_2)
- 10 ml/min argon (Ar)
- 0.1 s contact time (CT)
- 1 residual block
- 850°C temperature
- Mn – $\text{Na}_2\text{WO}_4/\text{SiO}_2$ catalyst

Catalyst screening at low contact time

Considering all permutations of M1, M2, M3, and support (7,410 permutations)

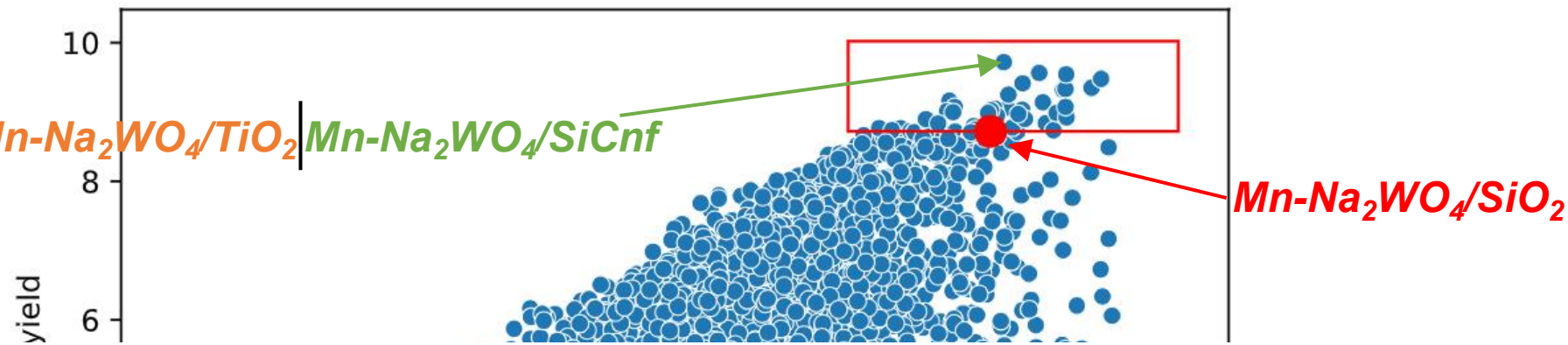


Operating Conditions:

- 10.5 ml/min methane (CH₄)
- 0.75 ml/min oxygen (O₂)
- 10 ml/min argon (Ar)
- 0.1 s contact time (CT)
- 1 residual block
- 850°C temperature
- Mn – Na₂WO₄/SiO₂ catalyst

Exploring reaction engineering solutions

Considering a two-bed reactor configuration of well-studied catalysts (3,481 combinations)



Operating Conditions:

- 10.5 ml/min methane (CH_4)
- 0.75 ml/min oxygen (O_2)
- 10 ml/min argon (Ar)
- 0.2 s contact time (CT)
- 1 block for each catalyst
- 850°C temperature

Catalyst 1	Catalyst 2	y_{C_2}	X_{CH_4}
$\text{Mn} - \text{Na}_2\text{WO}_4/\text{TiO}_2$	$\text{Mn} - \text{Na}_2\text{WO}_4/\text{SiCnf}$	9.72	11.65
$\text{Mn} - \text{Na}_2\text{WO}_4/\text{TiO}_2$	$\text{Mn} - \text{Na}_2\text{WO}_4/\text{SiC}$	9.56	12.02
$\text{Mn} - \text{Na}_2\text{WO}_4/\text{TiO}_2$	$\text{Mn} - \text{Na}_2\text{WO}_4/\text{SiO}_2$	9.54	12.31
$\text{Ti} - \text{NaWO}_4/\text{SiO}_2$	$\text{Mn} - \text{Na}_2\text{WO}_4/\text{SiCnf}$	9.41	11.85
$\text{Mn} - \text{Na}_2\text{WO}_4/\text{SiO}_2$	$\text{Mn} - \text{Na}_2\text{WO}_4/\text{SiO}_2$	8.72	11.51

Ongoing work: What if we learn a rate function?

Write the rate as a neural network

$$\frac{dF_i}{d\Omega} = r_i = f_{NN}(\textit{Catalyst}, \textit{Feed}, \textit{Temp})$$

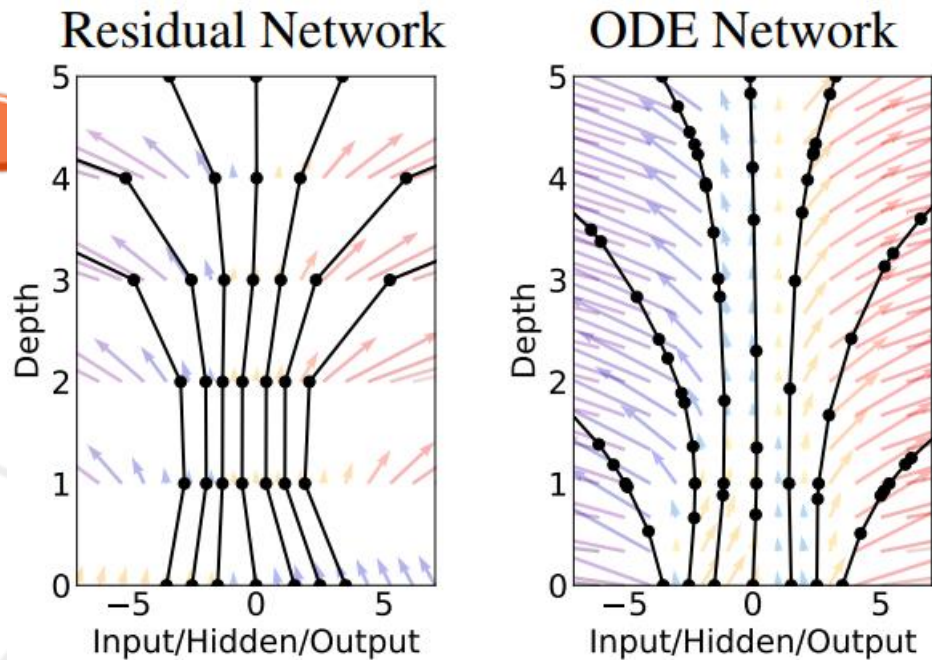
Include chemistry information

$$\frac{dF_i}{d\Omega} = Sr_j = S \times f_{NN}(\textit{Catalyst}, \textit{Feed}, \textit{Temp})$$

S : Stoichiometric matrix of the hypothesized reaction network

**Training the neural network requires integrating the ODE.
These are Neural ODEs!**

Neural ODEs



[1] Ricky T. Q. Chen, Yulia Rubanova, Jesse Bettencourt, and David Duvenaud. Neural ordinary differential equations. Advances in Neural Information Processing Systems, 2018.

Initial value problem (IVP) consists of:

- ODE
- Initial value, $\frac{dy}{dt} = f(t, y)$ $y(t_0) = y_0$

To solve an IVP using the default solver:

```
from torchdiffeq import odeint  
odeint(func, y0, t)
```

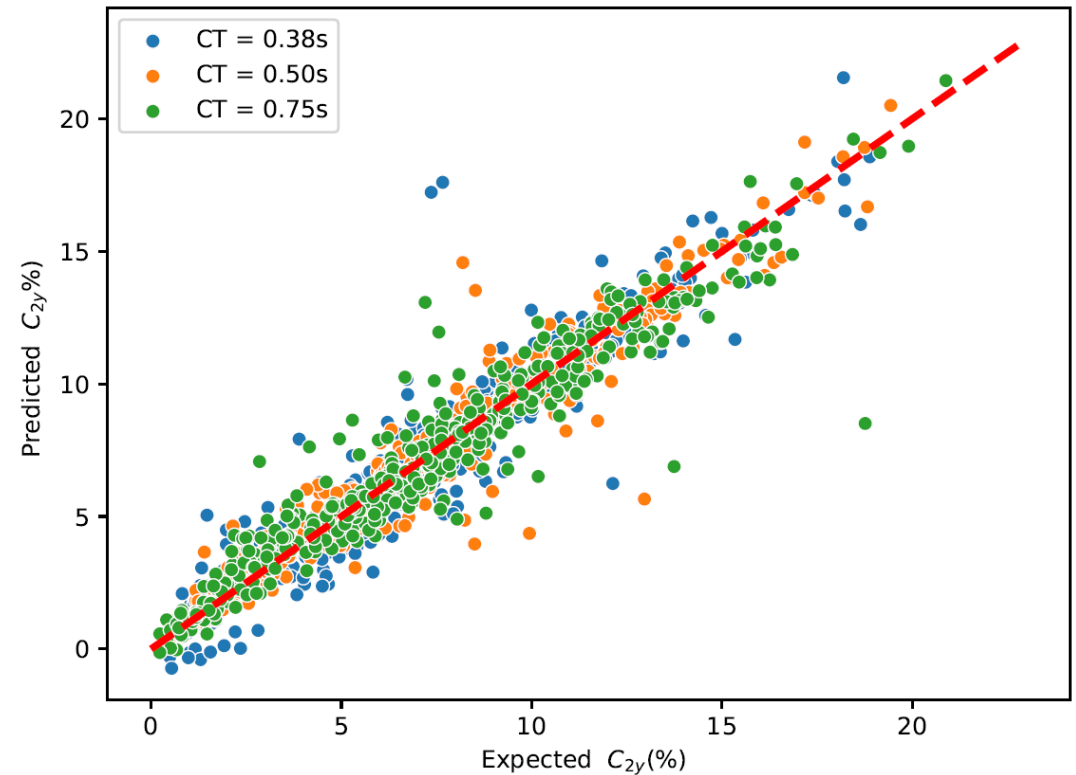
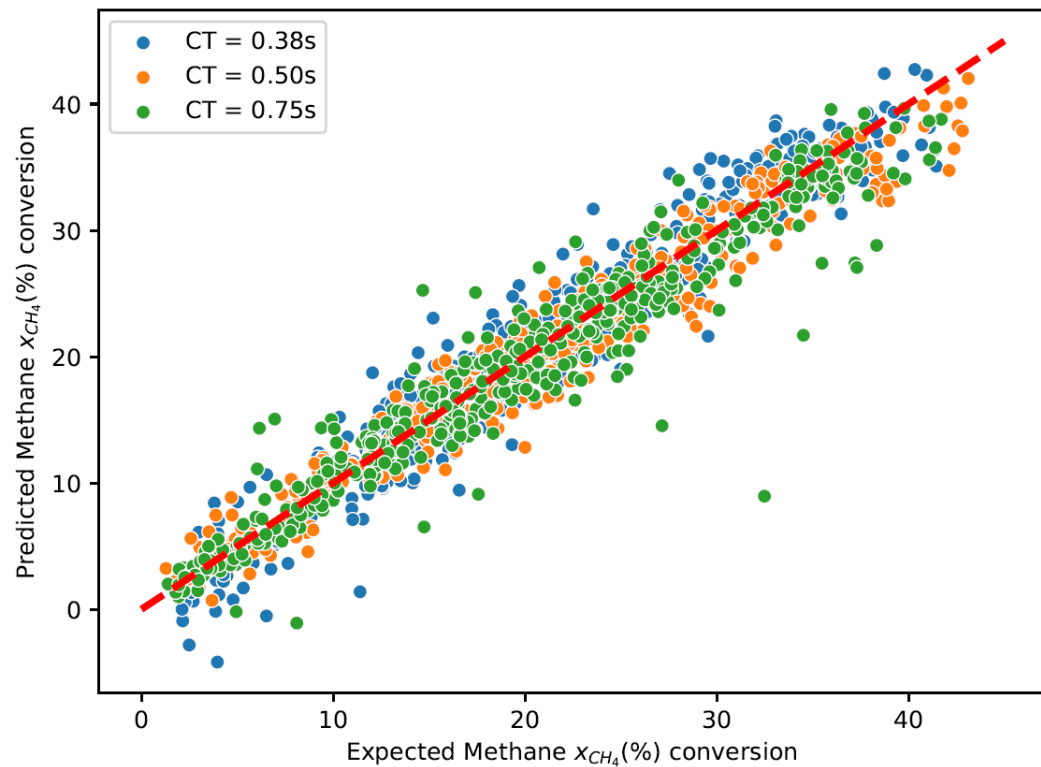
What are Neural ODEs?

- Neural ODEs are a new family of deep neural network models
- It combines neural networks with ODE to model dynamic systems
- The dynamics of hidden states are governed by ODEs
- ODEs are parametrized by neural networks
- These are continuous depth models that have constant cost memory

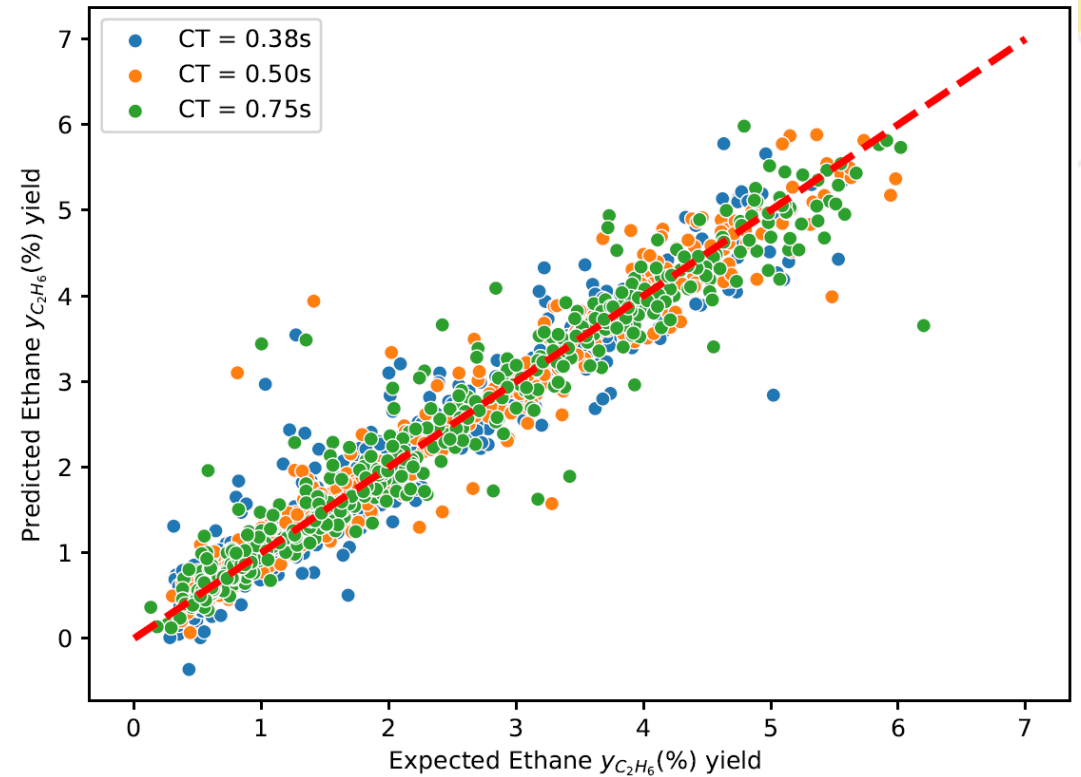
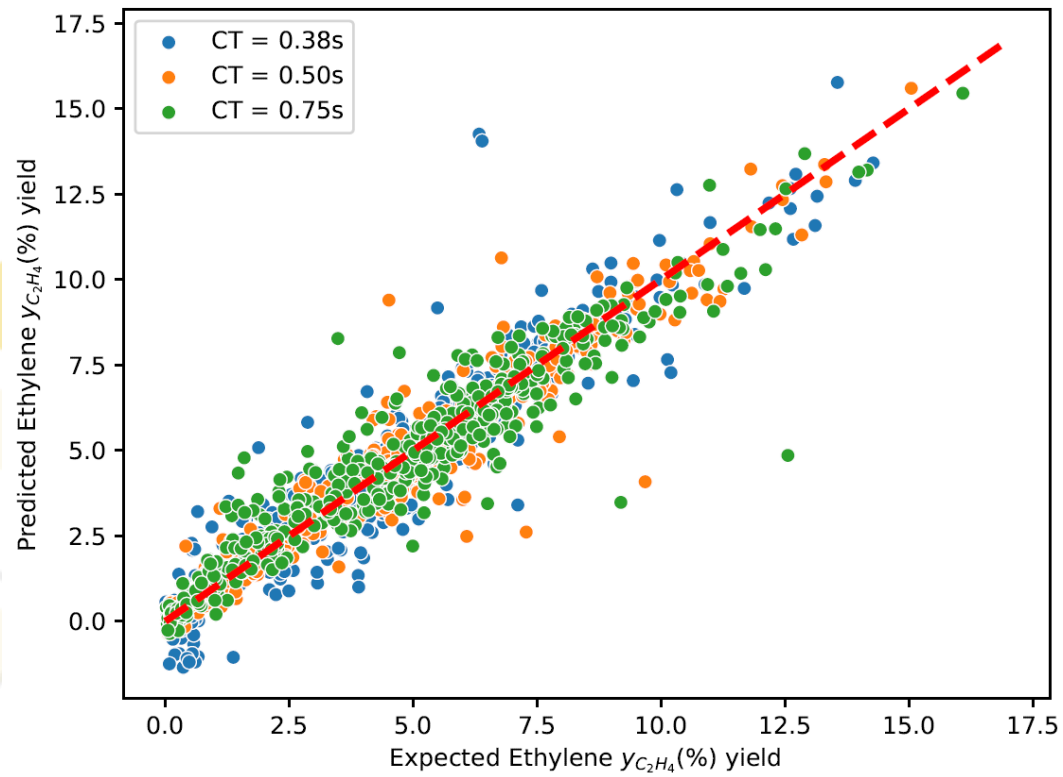
How to train Neural ODEs?

- We use torchdiffeq [1] library to train our model
- It provides ODE solvers implemented in PyTorch framework
- The goal of an ODE solver is to find a continuous trajectory satisfying the ODE that passes through the initial condition

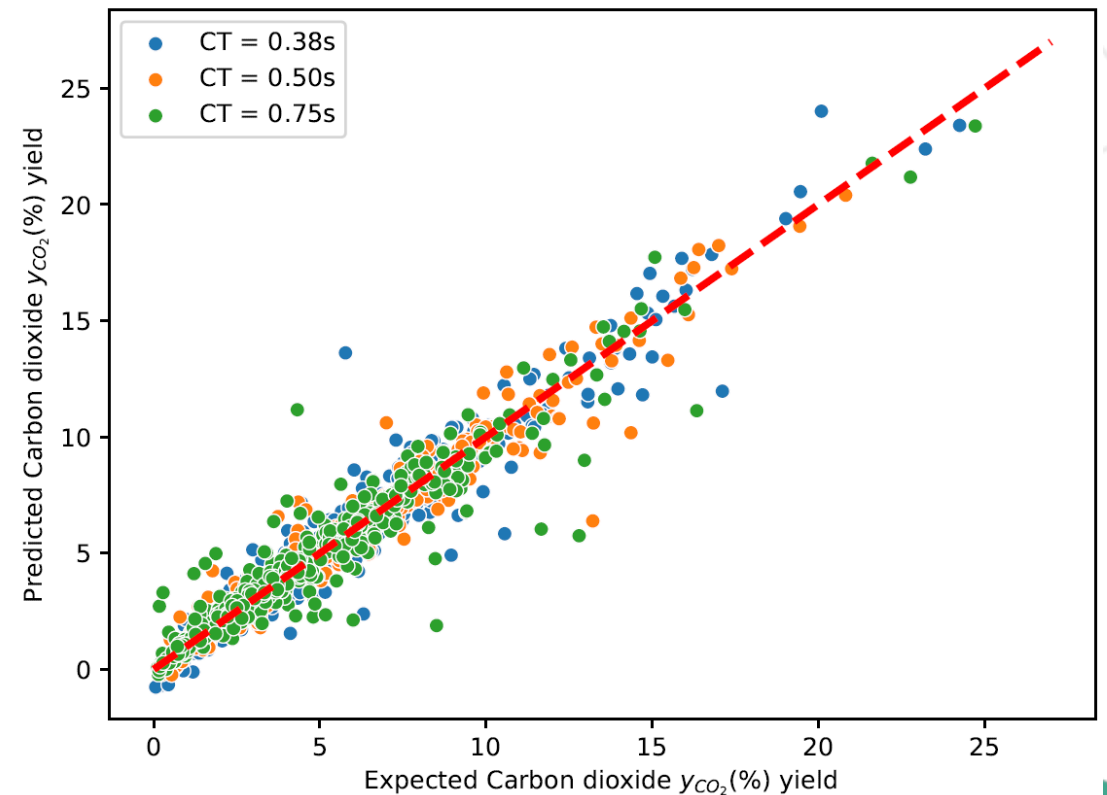
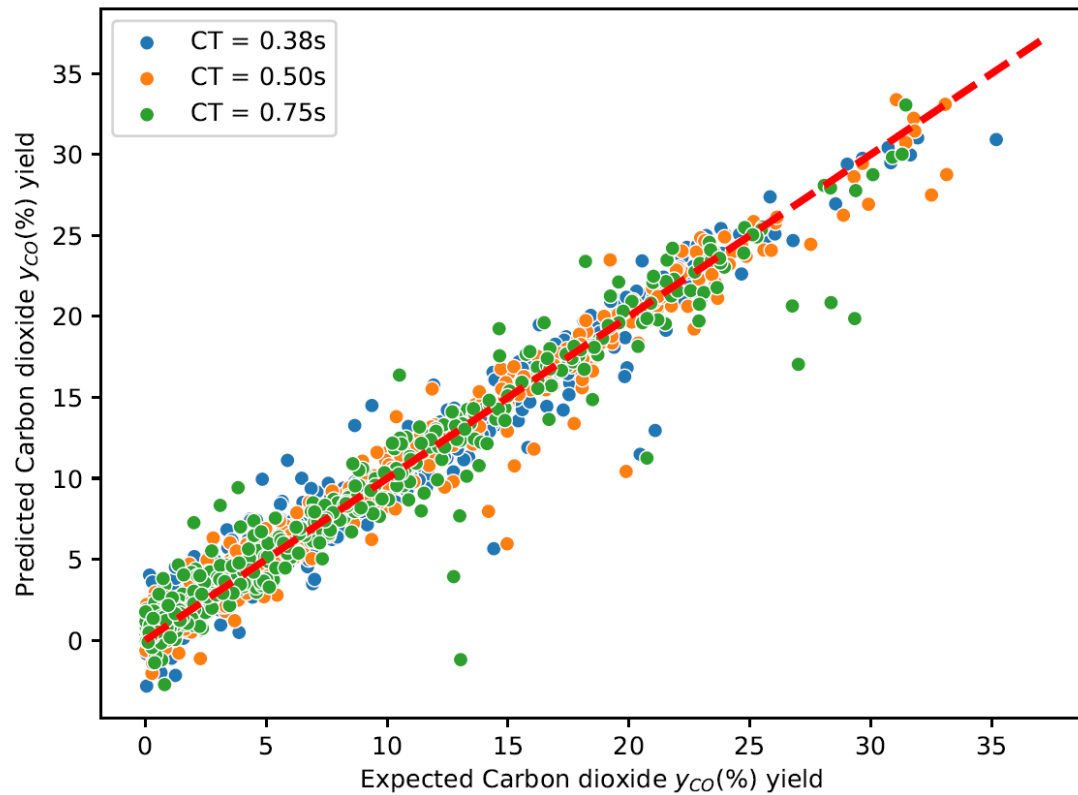
Preliminary Results using Neural ODEs to train the model : Parity Plots for Methane Conversion and C_2 Yield



Results: Parity Plots of Ethylene and Ethane yields



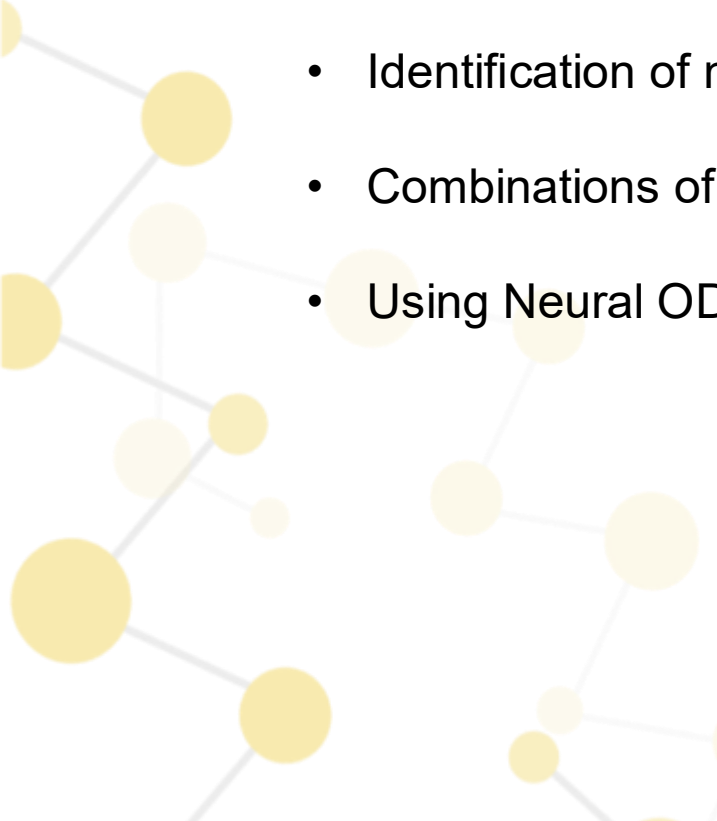
Results: Parity Plots of Carbon Monoxide and Dioxide yields





Summary



- We developed a DNN model for OCM to resemble the reactor design equation
 - This model allowed us to understand the variation of the reactant and product composition
 - Identification of new catalysts/ catalyst combinations
 - Combinations of known catalysts/ two bed catalysts combinations
 - Using Neural ODEs to capture design equation and chemistry information
- 
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Questions?
Thank You!

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